

Embodied functions and symbolic meaning of eyewear items in varied game environments

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ARTICLE INFO

Keywords:

Eyewear items
Functional roles
Game environment
Conventional meaning

ABSTRACT

This study examines the functional roles of eyewear items across three distinct game environments – PvX (hybrid PvP-PvE), PvP (player versus player), and PvE (player versus environment) – analyzing their relationship with real-world eyewear. Hierarchical clustering and text mining techniques are used to categorize eyewear items according to their functional roles in gaming environments. PvX eyewear items integrate PvP-oriented combat functions (e.g., dexterity, luck) with PvE-oriented conventional functions (e.g., collection, detection, intelligence), thereby enabling seamless adaptation to hybrid gameplay environments. PvP eyewear items support combat efficiency and display minimal association with the conventional meanings of real-world eyewear. PvE eyewear items support exploration and resource management and align with the conventional meanings of real-world eyewear. These functional meanings differ across PvX, PvP, and PvE environments as each emphasizes distinct gameplay priorities. These interactions demonstrate how eyewear items mediate the adaptation of real-world meanings within game environments. Therefore, game items are reconstructed by the interaction between game environments and conventional meanings. The meanings of game items are primarily shaped by the game's core content and environment, with real-world meanings playing a supplementary role.

1. Introduction

1.1. Background

Game genres and environments influence player behavior and how in-game items are experienced [1,2]. Game genres and environments shape gameplay structure and enhance player experiences by introducing specific challenges and objectives [3]. For example, MMORPGs (massively multiplayer online role-playing games) focus on character progression and large-scale interactions [4]. Simulation games emulate real-world experiences [5], while AOS (aon of strife) or MOBA (multiplayer online battle arena) emphasize real-time strategy and competitive team play [6]. Therefore, genre characteristics serve as key determinants of item types and functions in games.

Gaming environments are typically classified into four types: PvC (player versus computer), PvX (player versus x, encompassing both PvP and PvE), PvP (player versus player), and PvE (player versus environment) [7–11]. PvC refers to gameplay where players compete against

computer-controlled opponents, typically through structured modes such as challenges, training scenarios, or computer-driven battles. PvX represents hybrid gameplay that combines or alternates between PvP and PvE elements [12,13]. PvP focuses on direct player-versus-player combat, often within large-scale multiplayer environments. PvE involves interactions with the game world, including open-world exploration, NPC (non-player character) driven quests, and dungeon content. Each environment requires distinct item functions: combat support in PvP, exploration tools in PvE, and, in PvX, versatile items that adapt to diverse scenarios.

Game items functions are shaped by gameplay context, reflecting each environment's demands [14,15]. PvX environments, integrating both PvP and PvE aspects, demand versatile items capable of adapting to various gameplay scenarios. In PvP environments, items that enhance combat and competitiveness are essential. PvE environments prioritize items that support exploration and narrative progression. This classification underpins the strategies and utilization of game items across different contexts. [16,17].

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<https://doi.org/10.1016/j.entcom.2025.101047>

Received 23 June 2025; Received in revised form 29 September 2025; Accepted 28 October 2025

Available online 31 October 2025

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Beyond their functional roles, game items also carry symbolic meaning, often reflecting real-world associations and the public's imagination [18,19]. These symbolic meanings are not only embedded within in-game items but are also reinterpreted and adapted to fit different gaming environments, extending beyond their original functions [20]. Eyewear items, conventionally regarded as fashion accessories, hold significant potential to influence player experiences across gaming environments. In reality, glasses serve dual purposes as both fashion items and medical devices for vision correction, allowing them to seamlessly integrate into diverse gameplay scenarios by addressing both visual enhancement and player expression. These functional roles can be reflected in their games, blending aesthetic appeal with functional utility [21–23].

To explore these conventional meanings, hierarchical clustering analysis offers a powerful method for categorizing game item functions across different environments [24,25]. By quantifying the similarities and differences between clusters through hierarchical distances, this method enables a deeper understanding of how distinct item properties align with specific gaming environments [26,27]. Larger inter-cluster distances indicate that the clusters are independent and possess unique characteristics, reflecting distinct strategies. Conversely, shorter distances suggest that the clusters share similar attributes and are likely to be used in related contexts [28,29]. Thus, hierarchical clustering analysis proves to be an effective tool for identifying unique patterns and relationships within the functions of eyewear items.

1.2. Objectives

This study aims to analyze the functions of eyewear items used in various gaming environments and explore how these functions relate to eyewear in real life.

2. Methods

2.1. Data collection

Eyewear item functions were collected by searching for the keyword “eyewear” within a prominent gaming community, INVEN [31]. The data collection was conducted from July 29 to August 9, 2024. The search yielded a total of 146 eyewear items across 12 games, including ‘Uncharted Waters (The Age of Exploration)’, ‘Black Desert’, ‘World of Warcraft’, ‘Lineage 2’, ‘Maple Story’, ‘Lost Ark’, ‘Dungeon & Fighter (Dungeon Fighter Online)’, ‘Aion’, ‘Blade & Soul’, ‘Don’t Starve’, ‘Project Zomboid’, and ‘Cyphers’.

The decision to focus on online games featured in INVEN was driven by the significant influence of online games compared to video games. Online games encompass larger player bases and foster dynamic interactions within the player community through multiplayer environments, making them ideal for studying the use of in-game items [32]. INVEN, as one of the most active gaming communities, provides a rich database of player-generated content and detailed item information, ensuring a reliable and comprehensive dataset for analysis [33].

2.2. Data preprocessing

In this study, item function data were preprocessed before analysis. Through the preprocessing process, item functions from different games were standardized. For item functions, the integration process was centered on Aion, a game that encompasses both PvP and PvE attributes, making it highly representative of PvX games [34]. Aion has been

previously analyzed through text-mining methodologies, further validating its suitability for this study [35].

2.3. Data analysis

The collected eyewear items were analyzed to classify their characteristics, focusing on their functions, and to compare their roles across different gaming environments. The functions of game items were selected as the subject of analysis because they are universally applied across global servers [36].

The text data were preprocessed to ensure analytical compatibility, and hierarchical clustering was visualized using the orange3 software [37,38]. This approach enabled the identification of relationships between the distinctive functions of eyewear items across various gaming environments [39–44]. To interpret the hierarchical clustering results, hierarchical distances were incorporated into the semantic network analysis.

In Orange3, the distance between clusters is calculated using ‘euclidean distance’ based on ‘average inter-cluster dissimilarity’, with the maximum distance between groups standardized to 100 % [45,46]. Additionally, by applying ‘pruning: 20 levels’, hierarchical clustering is refined by removing clusters beyond the 20th level, ensuring an optimal number of clusters for meaningful data segmentation and analysis.

The hierarchical distance visually represents the relationships between clusters, where clusters with shorter euclidean distances merge earlier (toward the right side) [47]. In cases where only one interpretive variable is used, the y-axis of the dendrogram corresponds to the variable, while the x-axis represents the similarity (or distance) between clusters. In contrast, clusters with larger euclidean distances merge later (toward the left side), indicating greater dissimilarity between them [48].

3. Results

3.1. General characteristics of eyewear items

The distribution by game environment showed that PvX included 4 games, PvP 2 games, and PvE 6 games, whereas the number of eyewear items reveals a different pattern, with PvX and PvP each featuring 60 items (41.10 %) and PvE containing only 26 items (17.80 %) (Table 1). In terms of game genres, there were 8 MMORPGs, 3 simulation games, and 1 AOS (MOBA) game. Regarding item classification, 80 items were associated with MMORPGs, 58 items with AOS (MOBA), and 6 items with simulation games. The collected eyewear items showed that Cyphers had the highest number, contributing 58 items (39.73 %), followed by Aion with 44 items (30.14 %). Most other games featured fewer than 10 eyewear items.

Table 1
Classification results of eyewear items.

Game environment	Game genre	Game title	n (%)
PvX	MMORPG	Aion	44 (30.14 %)
		Black Desert	10 (6.85 %)
		World of Warcraft	4 (2.74 %)
		Blade & Soul	2 (1.37 %)
PvP	AOS(MOBA)	Cyphers	58 (39.73 %)
PvE	MMORPG	Lineage 2	2 (1.37 %)
	MMORPG	Maple Story	14 (9.59 %)
		Lost Ark	3 (2.05 %)
	Simulation	Dungeon & Fighter	1 (0.69 %)
		Don't Starve	3 (2.05 %)
		Uncharted Waters	3 (2.05 %)
		Project Zomboid	2 (1.37 %)
Total		12	146 (100 %)

3.2. Eyewear item functions in PvX environments

The frequency analysis of the top 18 keywords in PvX eyewear items reveals a balanced focus on both defensive resilience and exploration potential to support players in hybrid environments that combine PvP and PvE elements (Table 2). ‘Health point (40 times)’ is the most frequently mentioned attribute, emphasizing its critical role in ensuring durability and survivability in versatile gameplay scenarios. ‘Evasion (39 times)’ and ‘physical defense (38 times)’ follow closely, highlighting the importance of avoiding damage and enhancing resistance to physical attacks in combat and encounters [49,50]. ‘Magic resistance (37 times)’, ‘spell debuff (17 times)’, and ‘magic nullification (2 times)’ emphasize the need for protection against magical threats, ensuring players are well-defended against diverse attack types [51]. ‘Critical (18 times)’, ‘increases (15 times)’, ‘boost (15 times)’, and ‘critical hits (3 times)’ reflect its significance in maximizing potential damage when opportunities arise [52].

The functions of PvX eyewear items can be visualized as four distinct clusters with varying degrees of overlap and separation (Fig. 1). Cluster 1 (C1) stands completely apart from the other clusters, indicating a high level of independence with little functional overlap [53]. Cluster 2 (C2) is closer to clusters 3 (C3) and 4 (C4), sharing some similarities but still maintaining distinct features, suggesting partial overlap with these clusters. C3 and C4 have the most significant overlap, indicating that they share many common functions and are closely related. The PvX clusters are further subdivided into smaller hierarchical levels. C1 and C2 are divided into three hierarchical sub-levels, while C3 and C4 are divided into two hierarchical sub-levels.

The functions of PvX eyewear items can be categorized into four clusters, integrating both offensive and defensive capabilities for hybrid PvP and PvE adaptability (Table 3). C1 enhances survivability attributes. C2 provides attribute resistance and stat boosts for versatile combat. C3 improves magic abilities and aerial mobility. C4 enhances offense and adaptability.

PvX eyewear clusters are further subdivided into hierarchical levels, reflecting their multifunctionality. Their features combine combat-oriented attributes with survivability and utility functions. Additionally, some functions, such as ‘intelligence’ in C3, align with the traditional perception of eyewear enhancing intellect. ‘Flight speed’ resembles eyewear used by pilots, while ‘underwater breathing’ mirrors the role of goggles in aquatic environments.

Synthesizing these analyses, the PvX environment prioritizes essential combat functions such as survivability, evasion, and physical defense, indicating that PvP content plays a central role in the core gameplay structure of MMORPGs. In contrast, in PvE content, the symbolic functions of eyewear become more prominent, representing

Table 2
Keywords of Functions in PvX Eyewear Items.

No.	Frequency	Word
1	40	Health point
2	39	Evasion
3	38	Physical defense
4	37	Magic resistance
5	18	Critical
6	17	Spell debuff
7	15	Increases
8	15	Boost
9	13	Wind resist
10	13	Water resist
11	13	Earth resist
12	9	Intelligence
13	9	Flight speed
14	8	Fire resist
15	3	Critical hit
16	2	Underwater breathing
17	2	Spells
18	2	Magic nullification

attributes such as intelligence, flight, or underwater breathing [54].

3.3. Eyewear item functions in PvP environments

The frequency analysis of the top 14 keywords in PvP eyewear items reveals a strong emphasis on offensive capabilities and tactical advantages (Table 4). These items primarily enhance ‘critical hit (83 times)’, ‘movement speed (9 times)’, and ‘physical power (9 times)’, optimizing combat efficiency and providing strategic benefits in competitive gameplay. ‘Vision (8 times)’ and ‘detection range (2 times)’ underscore the importance of situational awareness and enemy detection in PvP scenarios, aligning with the real-world function of eyewear in enhancing visibility.

The functions of PvP eyewear items are classified into four clusters based on their similarity (Fig. 2). The PvP C2 cluster is further subdivided into two hierarchical sub-levels.

PvP eyewear items are categorized into four clusters, each enhancing combat effectiveness through offensive or strategic advantages (Table 5). C1 focuses on enhancing attack power and enabling decisive strikes. C2 includes items that enhance mobility and tactical advantages. C3 specializes in effects tailored for combat against human-type enemies. C4 consists of items that support long-range attacks and precision strikes, incorporating stealth, extended detection or attack range, and toxic or biological status-effect-based attacks.

The PvP C2 cluster is further subdivided into two hierarchical sub-levels, distinguishing between ‘vision’ and combat-related attribute functions. These PvP eyewear items primarily focus on combat effectiveness, with minimal association with conventional eyewear imagery. Apart from ‘vision’ in C2 and ‘detection range’ in C4, their resemblance to traditional eyewear remains limited.

Synthesizing these analyses, the PvP environment emphasizes combat efficiency through functions such as attack power, critical hits, and mobility. These features have minimal association with the conventional meaning of real-world eyewear, such as vision correction and intellectual enhancement. In AOS (MOBA) and MMORPG games in particular, eyewear items are contextualized as function-driven combat equipment, prioritizing strategic utility over aesthetic or conventional meaning [55].

3.4. Eyewear item functions in PvE environments

The frequency analysis of the top 12 keywords in PvE eyewear items reveals a strong emphasis on enhancing character attributes and survivability in non-competitive environments (Table 6). These items primarily focus on attribute enhancement, exploration, and survivability, with ‘intelligence (12 times)’, ‘dexterity (10 times)’, and ‘luck (10 times)’ being key attributes that support resource management and long-term progression.

The functions of PvE eyewear items are classified into four clusters based on their similarity (Fig. 3). The PvE clusters are further subdivided into smaller hierarchical levels, with C2, C3, and C4 each divided into two hierarchical sub-levels.

PvE eyewear items are categorized into four clusters, each contributing to resource management and survivability (Table 7). C1 focuses on intelligence-based attributes. C2 includes items that improve various abilities. C3 specializes in facilitating resource gathering, exploration, and environmental awareness [56]. C4 encompasses items that enhance survival and defense elements, contributing to stable combat performance in PvE environments.

The PvE clusters are subdivided into smaller hierarchical levels, supporting balanced growth, resource gathering, and environmental awareness. These PvE eyewear items primarily support resource management and survivability, closely aligning with conventional eyewear imagery. ‘Intelligence’ in C1 may be associated with the perception of eyewear enhancing intellectual appeal, holding greater significance in PvE settings. C3’s ‘collection’ and ‘detection’ functions reinforce the

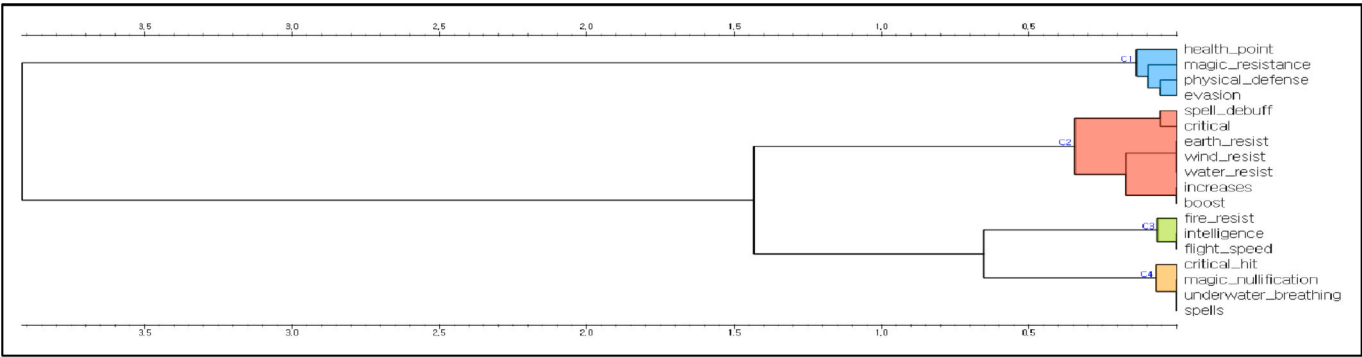


Fig. 1. Hierarchical Network of PvX Eyewear Item Functions.

Table 3
Hierarchical Cluster of Functions in PvX Eyewear Items.

Cluster	Keywords
C1	Health point, Magic resistance, Physical defense, Evasion
C2	Spell debuff, Critical, Earth resist, Wind resist, Water resist, Increases, Boost
C3	Fire resist, Intelligence, Flight speed
C4	Critical hit, Magic nullification, Underwater breathing, Spells

Table 4
Keywords of Functions in PvP Eyewear Items.

No.	Frequency	Word
1	83	Critical hit
2	9	Movement speed
3	9	Additional
4	9	Physical power
5	8	Vision
6	3	Against
7	3	Humans
8	2	Bug discharge
9	2	Parasite discharge
10	2	Attack range
11	2	Viper
12	2	Detection range
13	2	Dust ball
14	2	Sneak army

traditional role of eyewear in enhancing vision and awareness [57], while ‘accuracy’ in C4 aligns with eyewear’s role in improving precision. Synthesizing these analyses, the PvE environment highlights functions such as intelligence, accuracy, and detection, which are closely aligned with the cognitive and visual enhancement roles of real-world

eyewear. These attributes are especially prominent in simulation genres and PvE-focused content within MMORPGs, such as dungeons and exploration, fostering both immersive gameplay and conventional meaning for players [58,59].

Table 5
Hierarchical Cluster of Functions in PvP Eyewear Items.

Cluster	Keywords
C1	Critical hit
C2	Vision, Physical power, Movement speed, Additional
C3	Against, Humans
C4	Sneak army, Dust ball, Detection range, Viper, Attack range, Bug discharge, Parasite discharge

Table 6
Keywords of Functions in PvE Eyewear Items.

No.	Frequency	Word
1	12	Intelligence
2	10	Dexterity
3	10	Luck
4	9	All attribute
5	5	Health point
6	4	Accuracy
7	4	Magic point
8	4	Physical defense
9	4	Magic defense
10	3	Detection
11	3	Vision
12	2	Collection

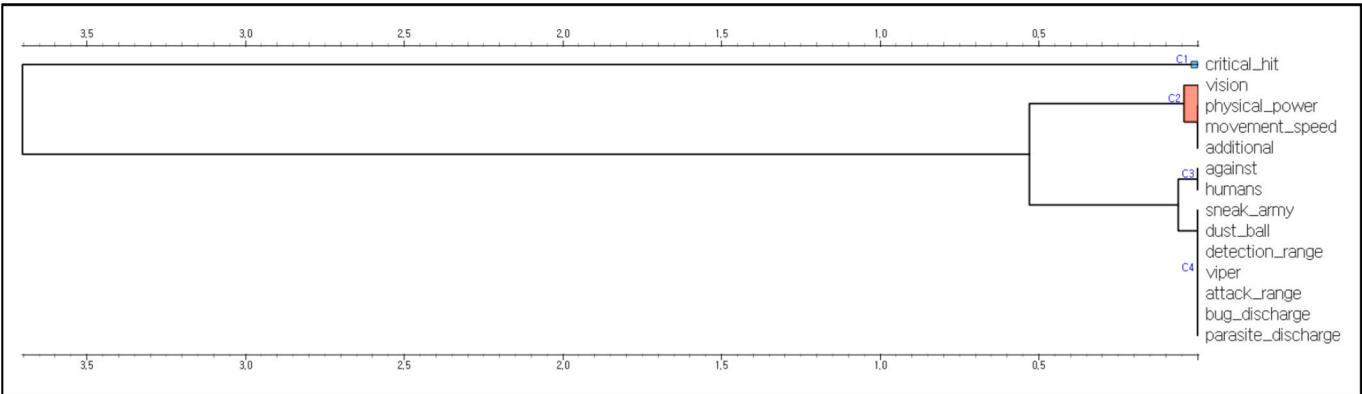


Fig. 2. Hierarchical Network of PvP Eyewear Item Functions.

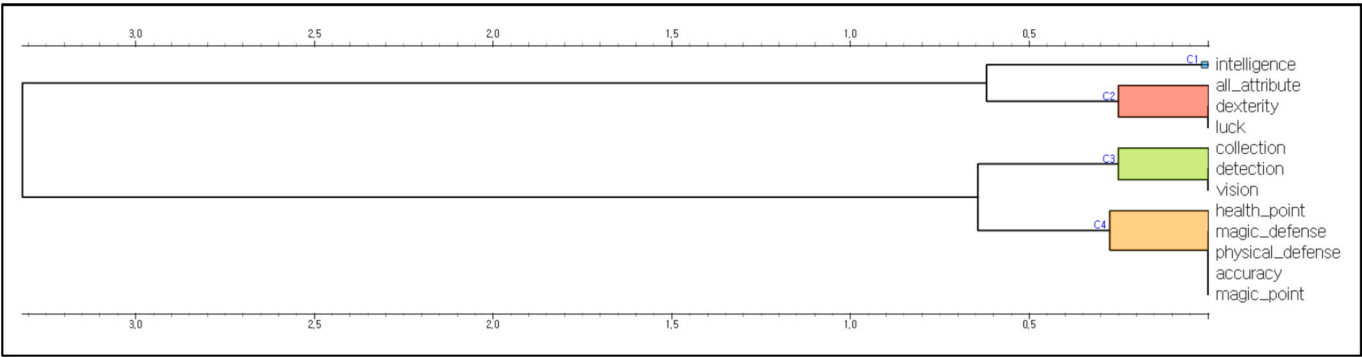


Fig. 3. Hierarchical Network of PvE Eyewear Item Functions.

Table 7
Hierarchical Cluster of Functions in PvE Eyewear Items.

Cluster	Keywords
C1	Intelligence
C2	All attribute, Dexterity, Luck
C3	Collection, Detection, Vision
C4	Health point, Magic defense, Physical defense, Accuracy, Magic point

4. Discussion

4.1. Key results

This study examined the functions of eyewear items across PvX, PvP, and PvE environments using hierarchical clustering analysis, revealing characteristics specific to each environment. PvX eyewear items support both combat and survival functions, making them versatile and adaptive. They include combat features (PvP) such as resilience, evasion, and exploration (PvE) related traits like conventional functions. This dual role indicates that PvX items are designed for hybrid environments.

PvP eyewear items strengthen offense and tactics for combat advantage. These items focus on enhancing attack power, mobility, and enemy detection. PvE eyewear items strengthen exploration and intelligence functions, supporting resource management, detection, and survivability. These items enhance vision, cognition, and perception.

4.2. Interpretation

Eyewear items were categorized into three types based on game environments: combat (PvP), exploration (PvE), and hybrid (PvX). While previous studies have treated game items as either functional or cosmetic [60], this study demonstrates that item meanings are shaped by different game environments.

In PvP environments, combat attributes such as critical hits, attack power, and mobility are emphasized. Eyewear items are used as a functional tool, with minimal association with vision correction or intellectual enhancement [6]. In PvE environments, attributes such as intelligence, exploration, and detection are emphasized. Eyewear items are used as cognitive and intellectual tools, reflecting their conventional meanings [5].

In PvX environments, the simultaneous demands of combat and exploration create a complex meaning for eyewear items [12,13]. MMORPGs (typically PvX environments) show a tendency toward environment differentiation between PvP and PvE. This shift is driven by factors such as player preferences, content specialization, and operational efficiency. As a result, PvP and PvE have evolved into distinct functional environments within the broader PvX structure [61].

Game environment shapes how eyewear items acquire real-world associations. MMORPGs (PvX) represent a hybrid system where functionality (PvP) and conventional meaning (PvE) interact. In AOS

(MOBA, PvP), eyewear items are paired with combat attributes, indirectly reflecting conventional meanings. In contrast, eyewear items in simulation games (PvE) emphasize exploration, survival, and cognition, more directly aligning with eyewear’s symbolic functions. These findings suggest that game item meanings are also shaped by the content [6,11,62,63].

Accordingly, previous player-centered research has tended to treat game items as conventional objects [64,65]. Studies on object imagery have focused on literary contexts, where meanings are shaped by cultural convention [66,67]. In contrast, game item imagery emerges through the interplay between conventional meanings and game-specific contexts. These items are structured by game environments, with real-world meanings playing a supplementary role.

5. Conclusion

This study explored the functional roles of eyewear items in online games, focusing on the intersection between game environments and real-world associations. Hierarchical clustering analysis identified three types of eyewear items, each reflecting the functional meanings of game environments: combat (PvP), exploration (PvE), and hybrid (PvX).

In PvP contexts, eyewear items serve practical functions, with minimal connection to conventional meanings like intelligence, vision, or professionalism. In contrast, PvE environments emphasize attributes such as intelligence, exploration, and detection, which closely reflect the conventional meaning of real-world eyewear. PvX environments, which combine combat and exploration, create an interaction between functionality and conventional meaning.

Therefore, game items are shaped by the interaction between game environments and conventional meanings. The meanings of game items are primarily shaped by the game’s core content and environment, with real-world meanings playing a supplementary role.

6. Limitation

This study analyzed the functions of eyewear items in games to derive characteristics specific to different gaming environments. It has the following limitations:

First, the study focused on data collected from popular online games in INVEN, which may have resulted in insufficient representation of certain genres or environments. This limitation could affect the generalizability of the findings. Second, the analysis was focused on the functions of the items, without incorporating qualitative data such as player immersion or preferences associated with the use of these items. Finally, the perception of eyewear item functions may vary depending on regional or cultural contexts.

CRedit authorship contribution statement

Samyoung Yu: Writing – review & editing, Writing – original draft,

Methodology, Investigation. **Jihye Ahn:** Writing – review & editing, Conceptualization. **Moonsung Choi:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Acknowledgments

The authors would like to thank Min Kim and Jamin Lee for their help with data collection.

Data availability

Data will be made available on request.

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